VASILIS PAPAGEORGIOU

Computer Sciences Ph.D. student
University of Wisconsin-Madison

vpapageorgio@wisc.edu ◇ Personal Website ◇ Google Scholar ◇ X ◇ Github

EDUCATION

University of Wisconsin-Madison (UW-Madison), *Madison, WI*
Ph.D. in Computer Sciences — Aug 2022 - present
Advisor: Dimitris Papailiopoulos

Technical University of Crete (TUC), *Chania, Greece*
MEng in Electrical and Computer Engineering — Sep 2016 - Oct 2021
Advisor: Aggelos Bletsas
Grade: 9.55/10 *(top 0.5% in school's history as of Oct 2021)*
Summa cum laude

RESEARCH INTERESTS

Reasoning models, RL post-training, test-time compute, continual learning, AI-agents.

RESEARCH EXPERIENCE

Graduate Research Assistant — Aug 2022 – present
University of Wisconsin–Madison, advised by Dimitris Papailiopoulos

- Research on reasoning models, RL post-training, and continual learning; currently leading a project on test-time compute in continual RL. Earlier: long-context retrieval and in-context learning.

Undergraduate Research Assistant — Dec 2020 – Oct 2021
Technical University of Crete

- Developed an algorithmic framework for distributed and asynchronous iterative algorithms in ambiently powered wireless sensor networks.

Electrical and Computer Engineering Intern — Jul 2019 - Jul 2019
Telecommunication Systems Research Institute, TUC

- Implemented a wireless communication link of two transmitting and a single receiving Software Defined Radios (SDR) using Matlab.
- Applied synchronization, channel estimation, detection techniques, and Alamouti's space-time block code.

INDUSTRY EXPERIENCE

Applied Scientist Intern — May 2025 - Oct 2025
AWS Agentic AI, Amazon.com – *Fundamental Research Team*

- Built a framework for utility-aware efficient reasoning that decides whether a reasoning LLM should attempt a problem at all.
- Trained lightweight predictors on the reasoning model's internal representations, optimized for a utility–cost trade-off between attempting and abstaining.

CONFERENCE PUBLICATIONS

- Zheyang Xiong, Ziyang Cai, John Cooper, Albert Ge, Vasilis Papageorgiou, Zack Sifakis, Angeliki Giannou, Ziqian Lin, Liu Yang, Saurabh Agarwal, Grigorios G Chrysos, Samet Oymak, Kangwook Lee, Dimitris Papailiopoulos. "Everything Everywhere All at Once: LLMs can In-Context Learn Multiple Tasks in Superposition". ICML 2025 (**Spotlight**).
- Zheyang Xiong, Vasilis Papageorgiou, Kangwook Lee, and Dimitris Papailiopoulos. "From Artificial Needles to Real Haystacks: Improving Retrieval Capabilities in LLMs by Finetuning on Synthetic Data." ICLR 2025.
- V. Papageorgiou, A. Nichoritis, P. Vasilakopoulos, G. Vougioukas and A. Bletsas, "Towards Ambiently Powered Inference on Wireless Sensor Networks: Asynchrony is the Key!," *2021 17th International Conference on Distributed Computing in Sensor Systems (DCOSS)*, Pafos, Cyprus, 2021, pp. 458-465 (received **best paper award**)

JOURNAL PUBLICATIONS

- V. Papageorgiou*, A. Nichoritis*, P. Vasilakopoulos*, G. Vougioukas and A. Bletsas, "Design and Implementation of Ambiently Powered Internet of Things-That-Think With Asynchronous Inference," in *IEEE Internet of Things Journal*, vol. 10, no. 17, pp. 15283-15300, 1 Sept.1, 2023, doi: 10.1109/JIOT.2023.3265423.

TEACHING ASSISTANTSHIP

CS 220 (Data Programming) Fall '24
University of Wisconsin-Madison

MATH 101 (Intro to Calculus and Analysis) Fall '19, Fall '20
Technical University of Crete

SCHOLARSHIPS & FELLOWSHIPS

Gerondelis Foundation Graduate Study Scholarship ($5,000) Nov 2023
Gerondelis Foundation

First-year CS Departmental Scholarship ($4,000) Aug 2022 - May 2023
University of Wisconsin-Madison

HONORS & AWARDS

Excellence Award for Undergraduate Studies Dec 2024
Technical Chamber of Greece

Summa cum laude Oct 2021
Technical University of Crete

2021 IEEE WCNEE Best Paper Award Jul 2021
IEEE

SOCIETIES & AFFILIATIONS

IEEE TUC Student Branch Oct 2018 - Oct 2021

SKILLS

Programming	Python, C, Java, Matlab
ML / DL	PyTorch, HuggingFace, Accelerate, TRL, VeRL, vLLM
Tools	Docker, Git, LaTeX
Languages	English (fluent), German, Greek (native)
Miscellaneous	Guitar